

CONTINUOUS CASTER OPERATIONS MANUAL

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Equipment Description

Continuous casters transform molten steel into semi-finished slabs through controlled solidification.

Normal Operating Parameters

- Mold Temperature: 700°C - 1000°C
- Cooling Water Flow: 250 - 450 m³/hr
- Oscillation System Vibration: 2 - 5 mm/s

Critical Thresholds

- Mold Temperature > 1100°C
- Oscillation Vibration > 8 mm/s
- Cooling Water Flow < 200 m³/hr

Common Failure Modes

1. Mold Copper Tube Wear
2. Oscillator Spring Failure
3. Cooling Water Restriction
4. Roll Bearing Wear

Troubleshooting Guide

High Mold Temperature

Possible Causes:

- Cooling system degradation
- Water flow reduction

Actions:

- Verify cooling pumps
- Inspect water circuits

Oscillation Alarm

Possible Causes:

- Spring fatigue
- Mechanical wear

Actions:

- Inspect oscillation mechanism
- Replace damaged springs

Maintenance Recommendations

Daily:

- Review mold temperatures

Weekly:

- Inspect cooling systems

Monthly:

- Oscillator inspection